

Generative Design of a Car Transmission System

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Abstract. The study focuses on optimizing the automobile power train in the commercial 4-wheelers sector, in order to obtain lightweight gears, which can be manufactured additively, without compromising on their structural properties. In general, heavy rotating masses such as gear wheels and shafts have a significant impact on fuel consumption. Current technology shows meager potential in lightweight gears mainly due to their manufacturability. With the advent of additive manufacturing, this problem can be tackled. A basic 5-gear transmission system gearbox adhering to the above was modeled using the CAD software SolidWorks, whose components have been illustrated so as to get an idea of what the system has to do, and thereby determine what all it needs to contain to do the same. Then, the model was exported to Fusion360, where the optimizing process called Generative Design had been applied. Generative Design is a paradigm of improved engineering design, transforming the engineering design and manufacturing process as it is today, with the advent of Industry 4.0. This is an AI-powered iterative computerized designing process wherein the requirements of the user and the operating environment of the model has to be adumbrated, upon which the software provides the user with a number of possible outcomes that would best solve the problem as described. Therefore, the operational environment of the gears had to be analyzed and the forces experienced by each one during its performance were calculated so as to feed into the software. The geometry of the design had to be demarcated into regions that were essential, and other regions that could be optimized by the program to provide better results. The final outcomes obtained were scrutinized and have been displayed at the end, giving the reader an impression of how a shape-optimized version of the usual common transmission system would appear.

INTRODUCTION

Transmission systems are one of the most important components in a vehicle. It transports power from the engine to the wheels. In a variety of car transmissions, some are automatic, while manual transmissions in stick shift cars require the driver to complete extra steps for the vehicle to operate effectively. When the vehicle is halted, the transmission disconnects the engine from the drive wheels in order to allow the idling condition of the engine when the wheels are not in motion. Transmissions also facilitate quick acceleration from rest and enable the engine to run more slowly to cut down on wear while the vehicle is driving at inconsistent speeds. Like any field of engineering, the area of transmission systems is also on the lookout for improvement. The development of electric and hybrid vehicles could potentially enable the elimination of mechanical transmission systems altogether and supplant them

with a programmed electrical transmission circuit, although it is believed to take a long time. Due to the rapid proliferation of a wide variety of technologies associated with AM, there is a lack of a comprehensive set of design principles, manufacturing guidelines, and standardization of best practices [1]. Moreover, the evolution of engineering design in order to utilise the possibilities afforded by AM and to manage the constraints associated with the technology has lagged behind [2]. Therefore, the existing transmission systems need to start acquiring upgrades in their form and functionality, to keep them competitive alongside novel technologies that are taking shape. The advent of 3D printing, or additive manufacturing, has opened up a world of possibilities for the designer [3].

Moreover, impediments due to climate change have been posing a challenge on the automobile industry due to the emission of harmful gases, as a result of which, light-weighted replacements for components of an automobile are in demand. Many of the constituents of emissions contribute to greenhouse gases, which absorb heat in the atmosphere, leading to increased temperatures and thus global warming [4]. Several parts of automobiles can be light weighted, especially rotating masses such as shafts and gears. However, there are limitations while modifying the gear design due to its manufacturability. Thales et al. say that while manufacturing gears, as a result of the elasticity of the equipment, current presses must operate with a minimum distance between punch and die to avoid tool failure when operating. This is with no working piece and enforcing some gear designs to have a more robust wall thickness than is usually needed due to this conceptual restriction. As a result, the wall thickness variation of the final product will be limited by this gap, which leads to a limited lightweight design [5].

An approach involving generative design for the additive manufacturing process is taken in this research. Design for Additive Manufacturing (DFAM) principles and technologies are applied in a multitude of domains, so as to provide solutions to alleviate problems especially involving limitations imposed by conventional designing and their manufacturing processes. The key aspect of DFAM is its ability to optimize models to a level of composition that is almost nearing perfection, in terms of reduction in mass, weight, material consumption, increase in precision, ergonomic compliance, energy storage, and transmission, wherever applicable [6].

Generative design is an iterative design exploration process powered by an Artificial Intelligence software. It provides a variety of designs based on the constraints given. According to Thomas Kuhn (Kuhn 1996), it offers a paradigm shift for the process of design and the expression of that process. For designers, it involves a reconsideration of the static artefact and the actions that manipulate it. It is closely tied to the general concept of synthesis, most viscerally apparent in nature and natural systems [7]. The method employed by additive manufacturing to construct objects customarily layer by layer, either by depositing material or by fusing liquid or solid particles by heating them using various sources, began as an option for designers seeking purely aesthetic or prototypical formulations [6].

METHODOLOGY

The main areas where optimization could be done with respect to the gears are their weight, and limitations posed in their manufacturability. These parameters are considered and modified to result in better outcomes, and the process involved therein has been elucidated. In brief, the transmission system works inside an automobile as follows. The engine runs on fuel and the combustion process creates mechanical torque, which is transmitted to the input gear, which passes it on to the input layshaft gear, which is keyed into the layshaft. There are five gears on the layshaft, which are coupled with corresponding gears on the output shaft, which would provide 5 speeds for the vehicle. In the end, there is a reverse gear-pair, which is connected by means of an idler. The latter exists only to change the direction of motion of the output gear such that the vehicle moves in an opposite direction. The output shaft rotates at the speed of one gear at a time, which can be chosen by means of the selector fork in conjunction with the synchronizer hub, which is operated by a cam mechanism. The entire gearbox lies within the gear case, which is attached to the shaft by means of bearings. The gears were modeled using SolidWorks 2020 software. They were created by utilizing basic formulas that relate various gear parameters such as the module, pressure angle, number of teeth, face width, and bore diameter, and used them as input. The required gear profiles were created by applying various pertinent combinations of the aforementioned parameters. The important design parameters for obtaining the gears are tabulated below.

TABLE 1. *Gear Parameters*

Gear	Module	Number of Teeth	Pressure Angle (degrees)	Face Width (mm)	Nominal Shaft Diameter (mm)
Input Gear	1.375	28	14.5	32	36
Input	1.375	42	14.5	32	68
Layshaft Gear					
Layshaft Gear 1	1.5	24	14.5	32	76
Layshaft Gear 2	1.5	28	14.5	32	76
Layshaft Gear 3	1.5	32	14.5	32	112
Layshaft Gear 4	1.5	32	14.5	32	128
Layshaft Gear 5	1.5	32	14.5	32	144
Reverse Gear		32	14.5	32	44
Layshaft					
Idler	1.5				36
Output Gear 1	1.5	32	14.5	32	112
Output Gear 2	1.5	32	14.5	32	112
Output Gear 3	1.5	32	14.5	32	112
Output Gear 4	1.5	32	14.5	32	112
Output Gear 5	1.5	32	14.5	32	112
Reverse Gear		32	14.5	32	104
Output					

Load Calculations

Generative design requires the load conditions experienced by the body of interest to be modeled into the software, with the level of accuracy involved being directly related to the potential of the obtained output towards surviving the operation environment. The loads to be withstood by a gear in operation could be chiefly described as a moment and the associated values had to be fed into the software. This value in the case of every gear had been calculated based on the model presented below. The calculations were made considering a common commercial 4-wheeler scenario of power 150 KW from the engine and a minimum speed of 800 rpm. The diagram with respect to these calculations is shown in Figure 3.

For the input layshaft and input gear,

$$n_p = 800 \text{ rpm, power} = 150 \text{ kW}$$

$$n_g = (z_p n_p)/z_g = 533.34 \text{ rpm} \quad (1)$$

$$M_t = (60 \times 10^6 \times \text{power})/(2\pi n_p) = 1790493.11 \text{ Nmm} = 1.79 \text{ Nm} \quad (2)$$

$$d_1 = mz'_1 = 38.5 \text{ mm}$$

$$P_t = 2M_t/d_1 = 93012.62 \text{ N} \quad (3)$$

$$P_r = P_t (\tan \alpha) = 24054.699 \text{ N} \quad (4)$$

Now, calculating for layshaft gear 1,

$$n_p = 533.34 \text{ rpm, } m = 1.5$$

$$z_g = z_7 = 42, z_1 = 24 = z_p, d = mz_1$$

$$(M_t)_1 = (60 \times 10^6 \times \text{power})/(2\pi n_p) = 2685739.665 \text{ Nmm} \quad (5)$$

$$(P_t)_1 = 2M_t/d = 149207.76 \text{ N} \quad (6)$$

$$(P_r)_1 = 38587.75 \text{ N}$$

$$(n_g)_7 = 304.76 \text{ rpm}$$

$$(M_t)_7 = (M_t)_1 \times (z_1/z_7) = 1534708.38 \text{ Nmm}$$

Repeating the calculations for layshaft gear 2

$$z_p = z_2 = 28, m = 1.5, n_p = 533.33 \text{ rpm}$$

$$z_g = z_8 = 38, \alpha = 14.5, d_p = mz_2 = 42 \text{ mm}$$

$$(M_t)_2 = 2685740 \text{ Nmm}$$

$$(P_t)_2 = 2M_t/d_2 = 127892.365 \text{ N} \quad (7)$$

$$(P_r)_2 = P_t(\tan\alpha) = 33075.214 \text{ N} \quad (8)$$

$$(n_g)_8 = 392.736 \text{ rpm}$$

$$(M_t)_8 = (M_t)_2 \times (z_2/z_8) = 1979000 \text{ Nmm} \quad (9)$$

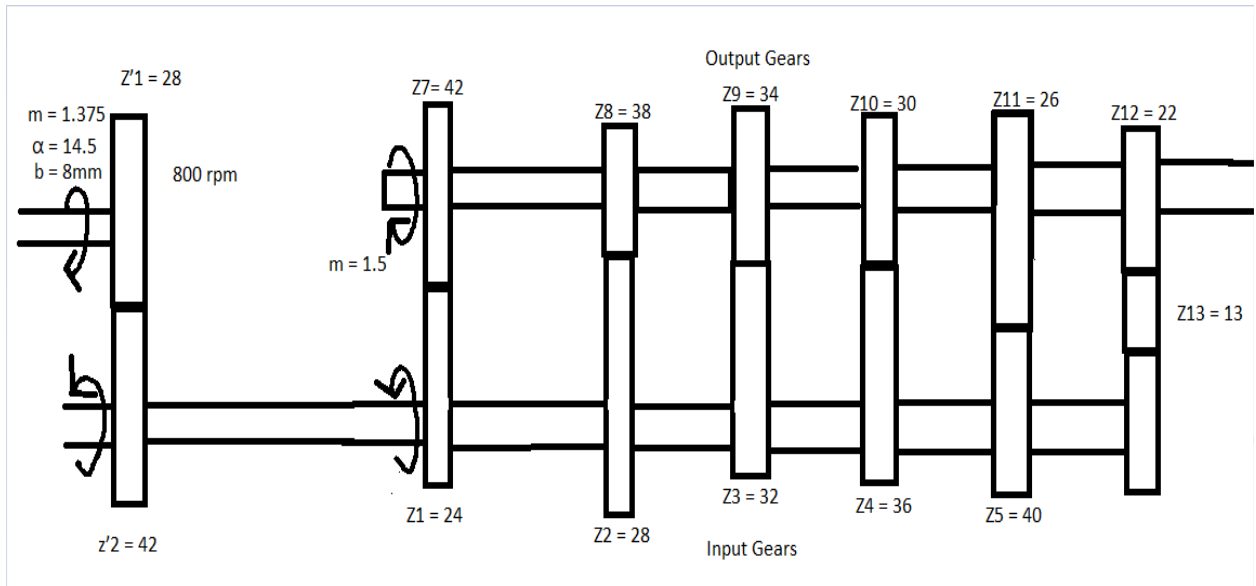
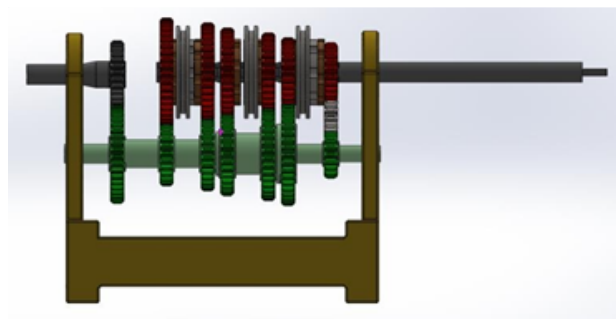
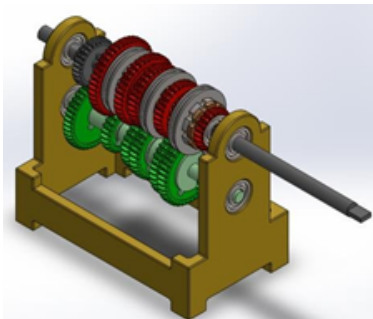


FIGURE 1. Load Diagram of the Gear Transmission System

Initial Design

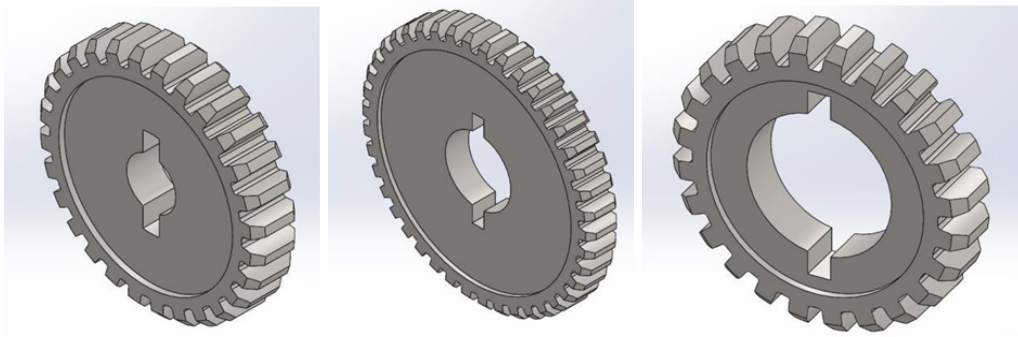
A model of a typical transmission system that could be employed in a car, with a 5-speed gear transmission, along with a reverse gear was designed. An illustration of the same design using SolidWorks CAD software is given in Figure 2. The design of each individual component is as represented in Figure 3.



a) Isometric View

b) Front View

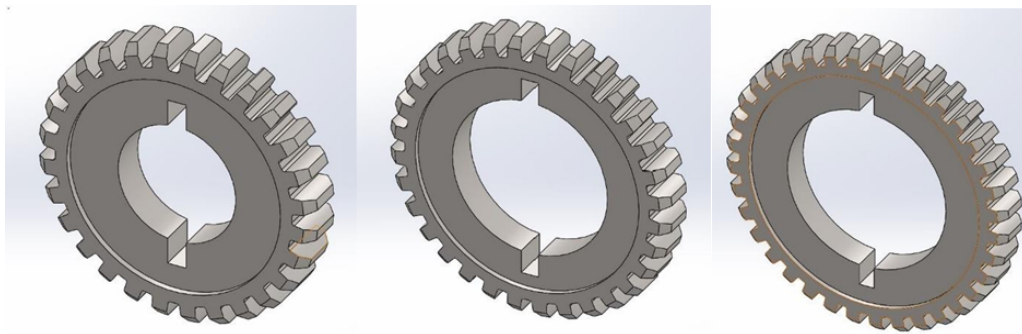
FIGURE 2. CAD model of gear transmission system



a) Input Gear

b) Input Layshaft Gear

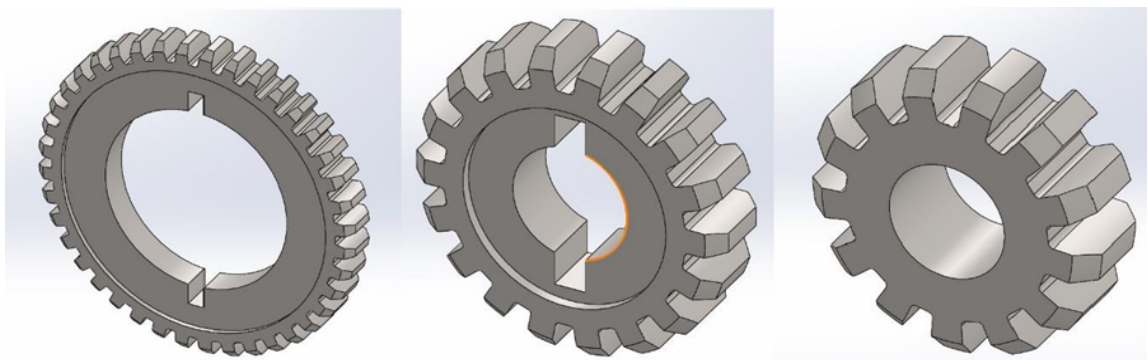
c) Layshaft Gear



d) Layshaft Gear 2

e) Layshaft Gear 3

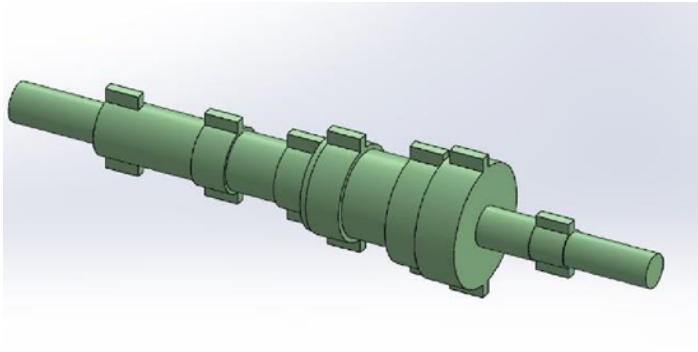
f) Layshaft Gear 4



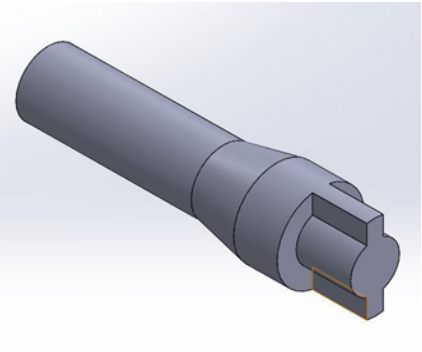
g) Layshaft Gear 5

h) Layshaft Reverse Gear

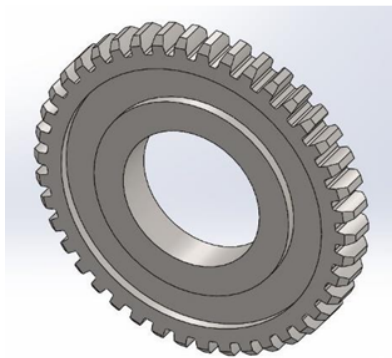
i) Idler



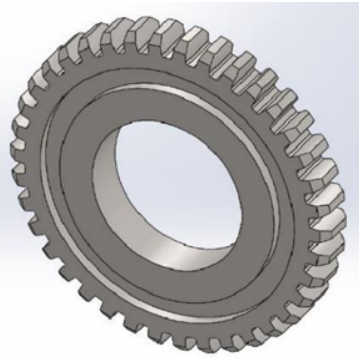
j) Layshaft



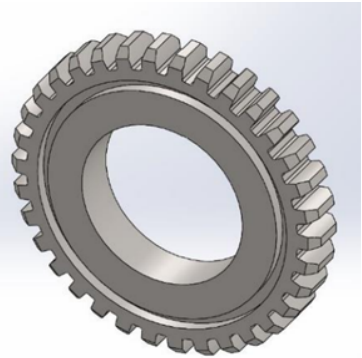
k) Input Gear Shaft



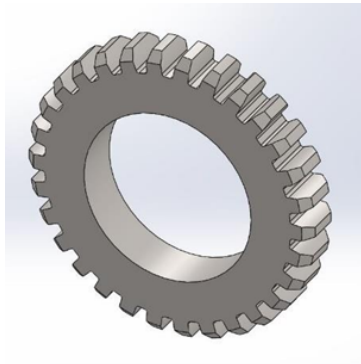
l) Output Gear 1



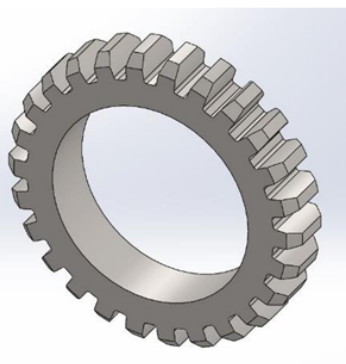
m) Output Gear 2



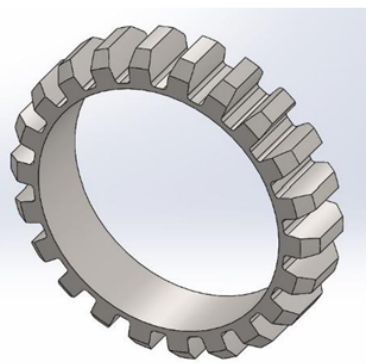
n) Output Gear 3



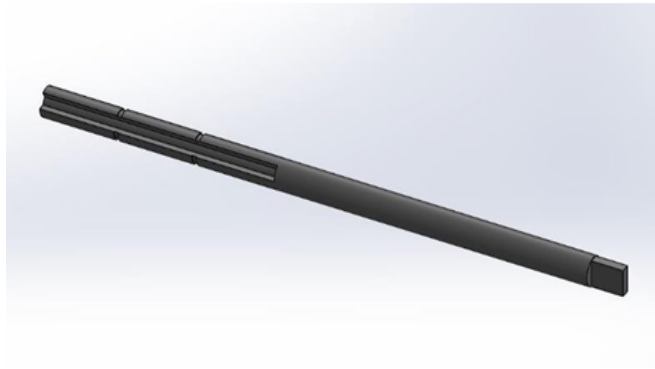
o) Output Gear 4



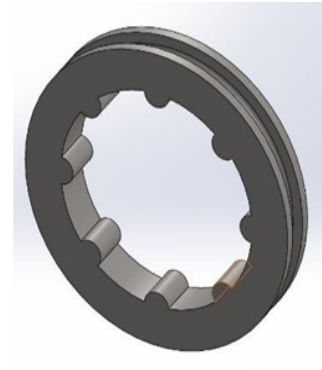
p) Output Gear 5



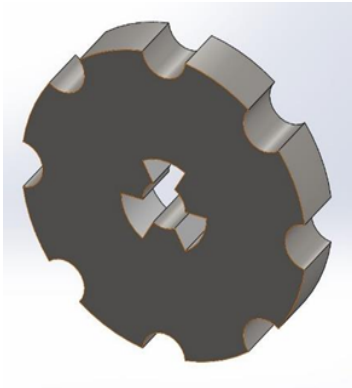
q) Output Reverse Gear



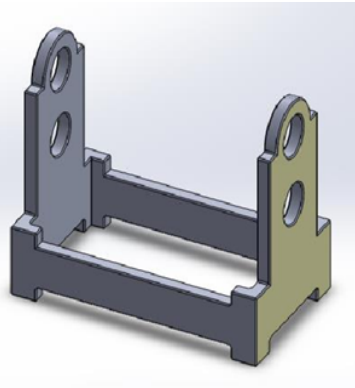
r) Output Shaft
Sleeve/Selector Fork



s) Sliding



t) Synchronizer Hub



u) Gear Case



v) Synchronizer Cone

FIGURE 3. *Individual components of the gear transmission system*

TABLE 2. *Gear load calculation results*

Gear	Output Speed (rpm)	Moment (N-mm)
Input Gear	533.33	298415.5183
Layshaft Gear 1	304.76	494741.5173
Layshaft Gear 2	392.736	631938.7448
Layshaft Gear 3	501.957	805721.8995
Layshaft Gear 4	639.996	1032976.794
Layshaft Gear 5	820	820642.6755
Layshaft Reverse Gear	436.36	298415.5183
Input Layshaft Gear	533.33	298415.5183
Output Gear 1	304.76	383677.095
Output Gear 2	392.736	494741.5173
Output Gear 3	501.957	631938.7448
Output Gear 4	639.996	805721.8995
Output Gear 5	820	1032976.794
Output Reverse Gear	436.36	820642.6755

GENERATIVE DESIGN

Generative Design is based on algorithmic and parametric modeling. It is a fast technique that is used to explore design possibilities or design optimizations [11]. For the performance of generative designing, there are a few precursors needed to be developed. Parametric and Generative Design techniques are more effective in decision-making processes, in the case of information and relationships that have to be defined [12]. Firstly, the loads experienced by the components in operational conditions are identified, so that they could be fed into the system so as to reinforce the material in those specific regions to withstand the load, and unnecessary material could be removed. Another precursor was the creation of obstacle geometry so that the software knows not to disturb certain essential regions, like the gear profile for example, and the slots and keyways made in the gears. This is made to ensure that the interaction of one component with its environment is not jeopardized as a result of this designing technique. This is done because each part of the transmission system would have a study performed on itself, independently. This activity is elaborated on, further in this paper.

Creation of Obstacle Geometry

Obstacle geometry is essentially the user indicating to the software that the associated areas are off-limits and could not be used for material addition or removal to optimize the design. Material added to these regions could prove detrimental to the system, and defeat the purpose of this design. Such additional geometry was needed to be created on the gears once they had been exported from SolidWorks to the Fusion360 workspace, so that the essential profiles could be preserved. Such geometry was created in the Autodesk Fusion360 CAD workspace and mounted onto the originally-designed assembly of the imported gears. This was done at a position that fulfilled the norms while being least intrusive to performance. At this juncture, the preparatory steps for generative design were in order [13]. Generative design has the unique feature of demarcating the components of a structural component study into the aforementioned parts not to be tampered with (obstacle regions), parts to be preserved (preserved regions), and the material that connects the preserved bodies. These parameters are set depending on the parts that require material addition or material removal as compared with the components that are uninvolved in the generative design. An encastre support is given along the internal circle of the gear. An obstacle body on the planes above and below the gears to ensure no material is added beyond this specific volume. Moreover, the tooth profiles of the gear are kept the same. The gear profiles and keyways for coupling with nearby shafts were preserved, and it was indicated to the software that the basic shape of the gear remains unchanged and material removal is done only between the shaft

hole and the internal circle. These are the critical regions for gear function. Two such setups are shown in Fig.4 and Fig.5 to indicate these regions, with respect to the input gear, and input layshaft gear.

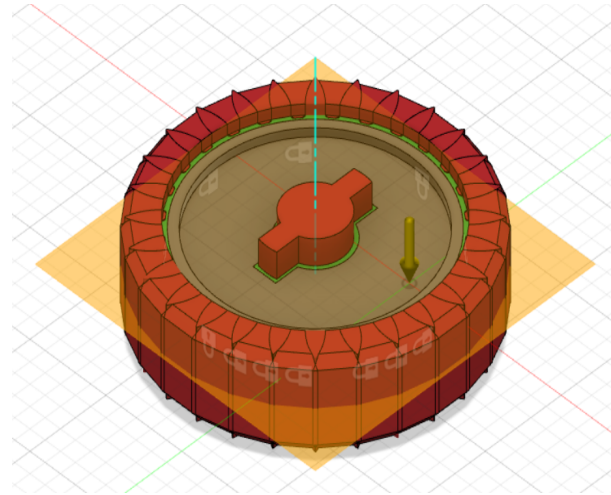
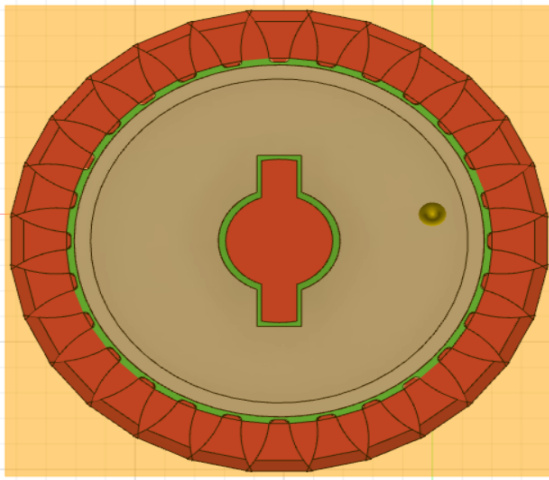


FIGURE 4. *Creation of obstacle geometry and preserved geometry for input gear*

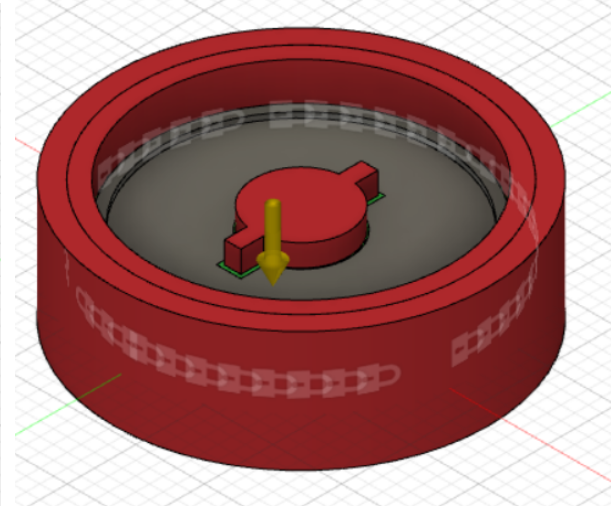
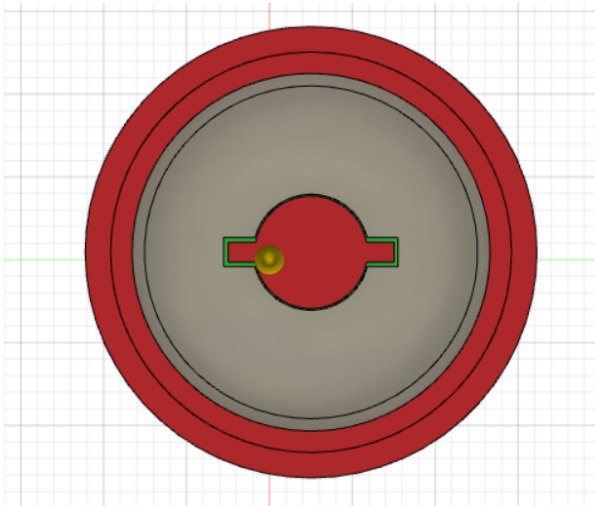


FIGURE 5. *Creation of obstacle geometry and preserved geometry for input layshaft gear*

Study Setup

Similar creation of geometry as mentioned above was done for the remainder of the gears. A factor of safety limit of 2 and a design objective of minimizing the mass was given. In order to obtain a multitude of designs, 3-axis milling, additive manufacturing, unrestricted and die casting were chosen for manufacturing methods, following which the generative design study set up was made to run.

MANUFACTURING METHODS AND MATERIALS

The manufacturing methods considered for the study are shown below. Based on the design objectives, the orientation of the geometry, constraints and loads given, the software picked out the best manufacturing method of its choice for this application, as displayed in Table 3.

TABLE 3. *Manufacturing methods and materials*

Manufacturing Methods	Materials
Additive Manufacturing	Aluminium AlSi10Mg
Milling	Aluminium 6061
Die Casting	Aluminium A356 T6

RESULTS

Subsequent to the generation of the study set up, 10 iterative solutions for each gear were obtained, giving a total of 140 outcomes, of which the most optimum results were selected manually, on the recommendation provided by the software, as shown in the graph in Fig.6. The gears were selected on the basis of mass reduction percentages, compactness of the design, aesthetics and the ability to withhold the maximum load. The solutions can be explored by means of graphs automatically generated by the software, which allow an efficient mapping of the solutions allowing the user to select which parameters to use on the axes and how to group the solution to identify general trends. [14]

The details of a single outcome is given in Table 4 explicitly and the most recommended outcomes by the software in Table 5. The gears were successfully and are displayed in Fig.7 and Fig.8.

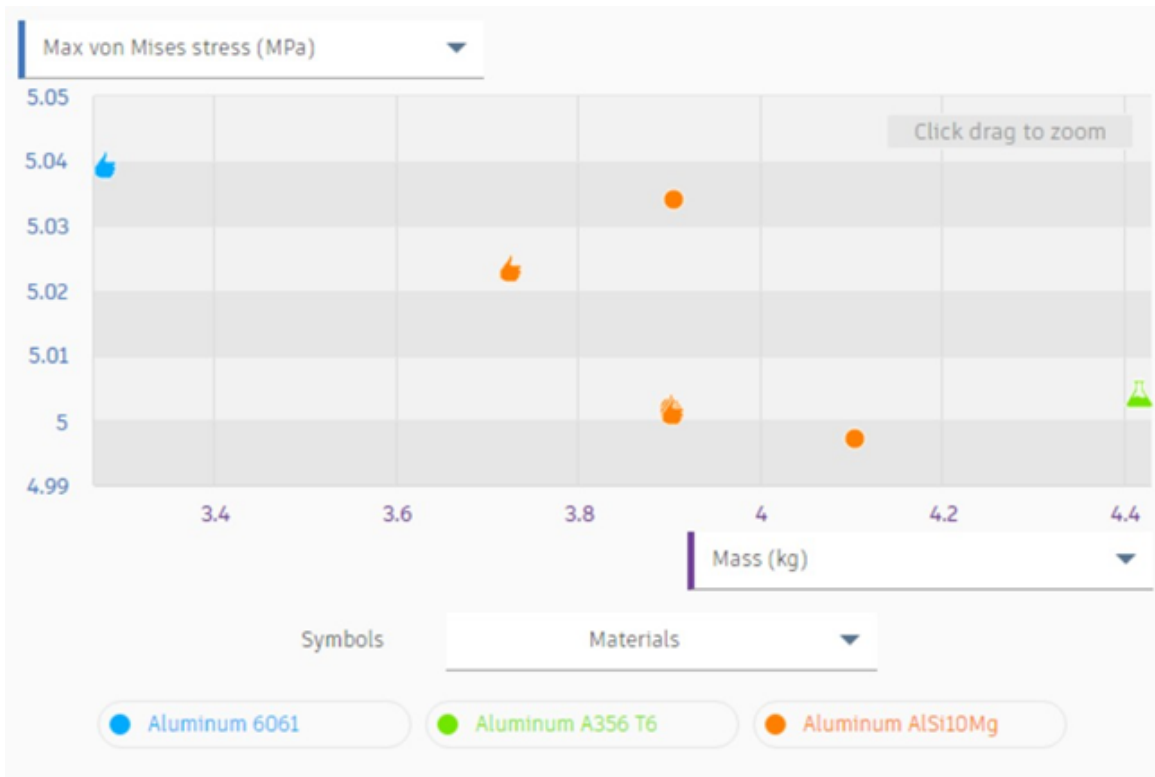


FIGURE 6. *Comparison of outcomes and their properties*

TABLE 4. Detailed properties view of a single outcome

Structural Study Outcome 1	Completed
Properties	Completed
Status Material	Aluminium 6061
Orientation	X+, Y+, Z+, X-, Y-, Z-
Manufacturing method	3-axis milling
Visual similarity	Unique
Production volume (pcs)	-
Piece part cost	
Range (USD)	-
Median (USD)	-
Fully burdened cost	
Range (USD)	-
Median (USD)	-
Volume (mm ³)	1.214e+6
Mass (kg)	3.278
Max von Mises stress	5
Factor of safety limit	2
Min factor of safety	54.57
Max displacement global (mm)	0

TABLE 5. Results of the outcomes based on Generative Design recommendations

Gear	Outcome Number Selected	Material	Manufacturing Method	Mass Reduction (%)	Von Mises Stress (MPa)
Input Gear	2	AlSi10Mg	Additive	56.6	40.7
Input	4	AlSi10Mg	Additive	33.3	45.1
Layshaft Gear					
Layshaft Gear 1	8	AlSi10Mg	Additive	63.3	44.7
Layshaft Gear 2	5	AlSi10Mg	Milling	57.4	45
Layshaft Gear 3	1	Al-6061	Additive	62.3	136.6
Layshaft Gear 4	5	AlSi10Mg	Additive	85.4	88.9
Layshaft Gear 5	6	AlSi10Mg	Additive	47.6	38.6
Reverse Gear	3	AlSi10Mg	Additive	44.7	38.8
Layshaft					
Output Gear 1	1	Al-6061	Milling	57.9	51.3
Output Gear 2	4	Al-A356 T6	Die Casting	87.2	57.3
Output Gear 3	5	AlSi10Mg	Additive	23.1	62.8
Output Gear 4	1	Al-6061	Additive	66.6	14.3
Output Gear 5	1	Al-6061	Milling	74.4	28.7
Reverse Gear	8	AlSi10Mg	Additive	67.7	23.5
Output					

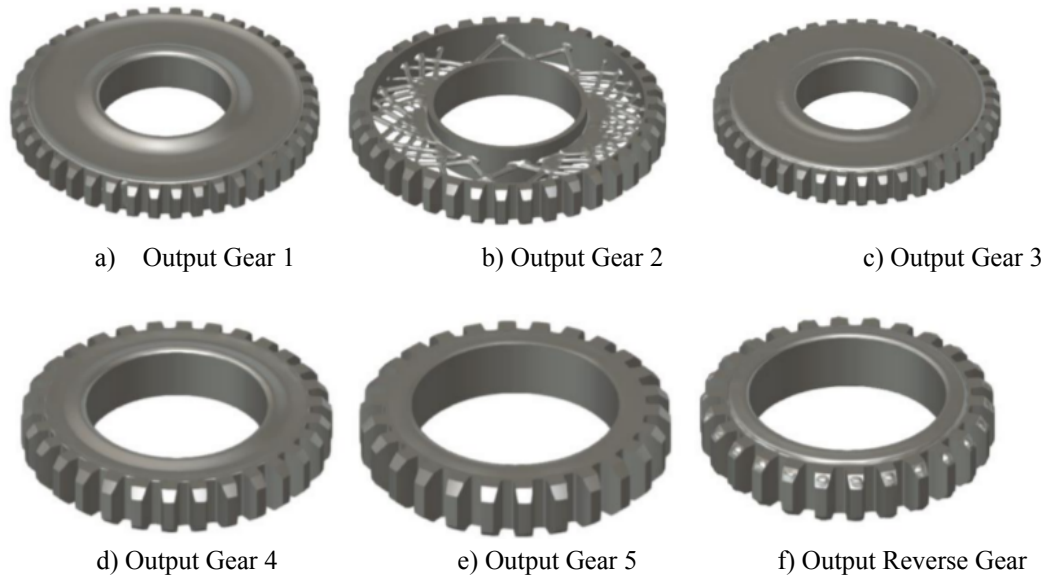


FIGURE 7. Generated outcomes of different output gears

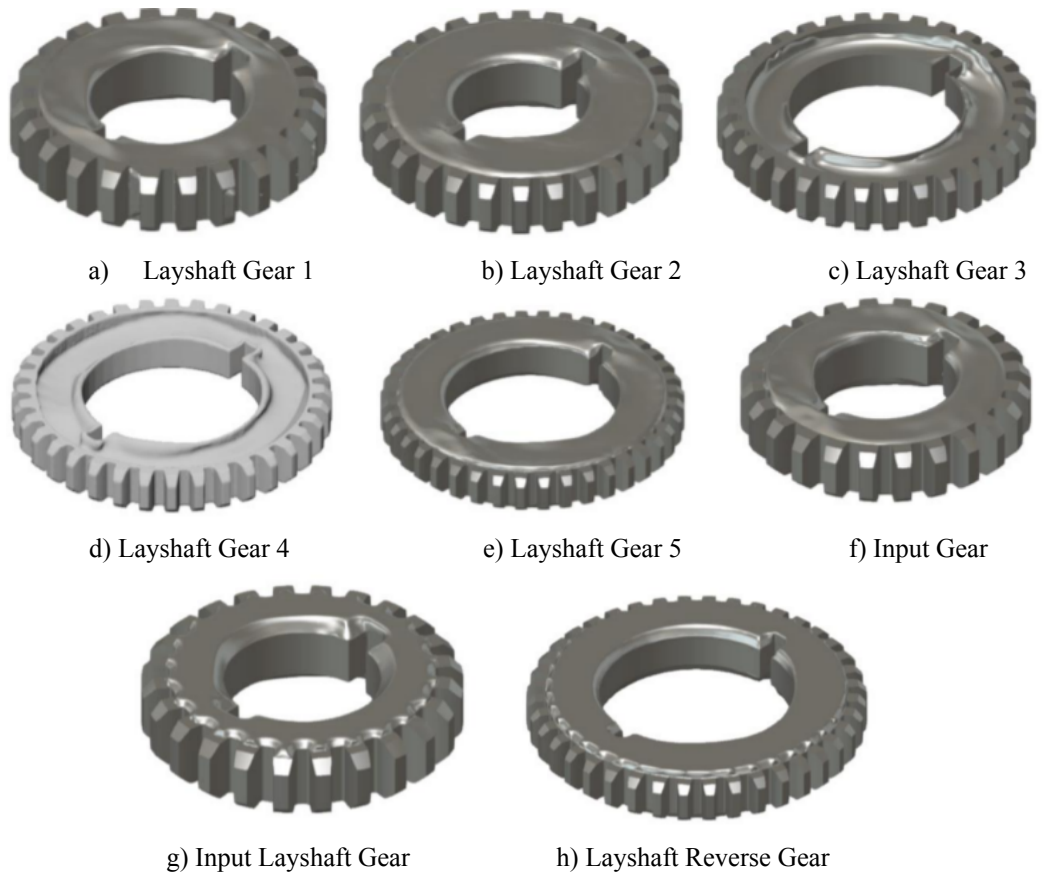


FIGURE 8. Generated outcomes of different input and layshaft gears

CONCLUSION

The results obtained at the end of generative design demonstrated a significant percentage of mass reduction, meaning that such designs would be able to satisfy the performance requirements posed to them, in case of their optimized versions just as well as the original ones. This substantiates the claim that automobile transmission systems have scope for such design changes leading to leaner and more efficient parts. Furthermore, 10 out of the 14 gears have additive manufacturing as the most recommended design. And this result could be a means for further impetus to be provided in the area of additively-manufactured transmission systems. This has been facilitated by the virtue of generative design. Moreover, the optimized gears can be assembled onto the original setup accompanied by the shafts and bearings. The gears could also be meshed into one another and run smoothly since the loads in operation had all been accounted for and the design would perform well in such conditions, provided they were manufactured efficiently as per the given choice of material and manufacturing method. Further work could be done in this regard using T-spline manipulation in Fusion360 to provide more symmetric models of the transmission elements, which would be very much conducive to the large-scale production of each design. Additionally, the replacement of aluminum with more additive manufacturing-friendly materials can be done to aid in bringing out better results to this avenue. Another aspect to ponder over is the optimization of other supportive elements in the transmission system i.e. bearings and synchronizer mechanisms, which have admittedly more complicated computation in terms of operational conditions. This study is aimed to serve as a precursor into a novel arena of generative design, and is hoped to bring out continuing works of the like.

ACKNOWLEDGMENTS

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